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# Microglia are Required for Developmental Specification of AgRP Innervation in the Hypothalamus of Offspring Exposed to Maternal High-Fat Diet During Lactation

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## eLife Assessment

The authors provide a **valuable** contribution by documenting the role of microglia in pruning the axon terminals of AgRP neurons. The analysis of microglial axonal pruning is **solid**; however, the analysis of the effects inhibiting microglia on subsequent food consumption is not fully complete.

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## Abstract

Agouti-related peptide (AgRP) neurons in the arcuate nucleus of the hypothalamus respond to multiple metabolic signals and distribute neuroendocrine information to other brain regions such as the paraventricular hypothalamic nucleus (PVH), which plays a central role in metabolic homeostasis. Neural projections from AgRP neurons to the PVH form during the postnatal lactational period in mice and these projections are reduced in offspring of dams that consumed a high-fat diet (HFD) during lactation (MHFD-L). Here we used immunohistochemistry to visualize microglial morphology in MHFD-L offspring and identified changes that were regionally localized to the PVH and appeared temporally restricted to the period when AgRP neurons innervate this region. In addition, axon labeling experiments revealed that microglia engulf AgRP terminals in the PVH, and that the density of AgRP innervation to the PVH in MHFD-L offspring may be dependent on microglia, because microglial depletion blocked the decrease in PVH AgRP innervation observed in MHFD-L offspring, as well as prevented the increased body weight exhibited at weaning. Together, these findings suggest that microglia are activated by exposure to MHFD-L and interact directly with AgRP axons during postnatal development to permanently alter innervation of the PVH, with implications for developmental programming of metabolic phenotype.

## Impact Statement

Microglia appear to play an essential role in specifying patterns of hypothalamic innervation during development in response to maternal HFD exposure, which may contribute to developmental programming of metabolic phenotype.

## 1 Introduction

Maternal nutritional status has a profound effect on the metabolic phenotype of offspring. Children born to obese mothers experience higher rates of obesity later in life, with accompanying comorbidities that negatively impact health and longevity (Stettler et al., 2005 [↗](#); Prospective Studies Collaboration 2009 [↗](#); Tamashiro & Moran 2010 [↗](#); Andersen et al., 2012 [↗](#)). Although this developmental programming of metabolic phenotype has been reproduced in several animal models (Samuelsson et al., 2008 [↗](#); Masuyama et al., 2014 [↗](#); García-Cáceres et al., 2019 [↗](#); Skowronski et al., 2023 [↗](#)), the underlying mechanisms remain poorly defined. In mouse models, maternal obesity during lactation, a time when offspring are dependent on milk from their mothers for nutrition, appears to be particularly impactful. These changes to metabolic phenotype are thought to be mediated by changes in the milk (Gorski et al., 2006 [↗](#); Vogt et al., 2014 [↗](#); Calvo-Lerma et al., 2022 [↗](#)) and occur without subsequent dietary challenge to the offspring themselves, suggesting that they are a consequence of developmental programming (Bolton et al., 2022 [↗](#); Skowronski et al., 2023 [↗](#)). Because neural circuits known to control body weight develop during the lactational period, they are vulnerable to a variety of environmental signals that may affect their organization and function (Horvath et al., 2010 [↗](#); Bouret et al., 2015 [↗](#); Elson & Simerly 2015 [↗](#); Zeltser 2018 [↗](#); Skowronski et al., 2023 [↗](#)).

AgRP neurons in the arcuate nucleus of the hypothalamus (ARH) function as “hunger neurons” that respond to key metabolic signals such as leptin, ghrelin, glucose and free fatty acids (Krashes et al., 2011 [↗](#); Betley et al., 2015 [↗](#); Chen et al., 2015 [↗](#); Sutton Hickey et al., 2023 [↗](#)), and they distribute this information to other regions associated with energy balance regulation (Simerly 2008 [↗](#); Zagmutt, et al., 2018 [↗](#)). Thus, the ability of AgRP neurons to influence other components of feeding circuitry is dependent on the formation of their neural connections, which develop primarily during the first 2 weeks of postnatal life (Bouret et al., 2004a [↗](#)). During development, AgRP axons extend from the ARH at postnatal day 4 (P4) and reach the paraventricular hypothalamic nucleus (PVH) between P8-P10. Leptin is required for normal targeting of AgRP axons to downstream regions, and in leptin-deficient mice both neuroanatomical and related physiological defects persist into adulthood (Bouret et al., 2004b [↗](#); Bouyer & Simerly 2013 [↗](#)). Maternal overnutrition affects formation of feeding circuits during postnatal life with concomitant dysregulation of body weight (Plagemann et al., 1992 [↗](#); Lippert & Bruning 2022 [↗](#); Skowronski et al., 2023 [↗](#)). Limiting HFD exposure of dams to the first 3 weeks of lactation (MHFD-L) causes suppression of neural projections from AgRP neurons to the PVH in offspring and is associated with increased body weight later in life (Vogt et al., 2014 [↗](#)). In fact, MHFD-L was more effective than prenatal maternal HFD exposure in causing body weight changes of adult offspring. MHFD-L did not alter cell number, peptidergic expression, or cellular activity of AgRP neurons in the ARH, suggesting that maternal nutritional status during lactation is particularly important for the establishment of neural connections related to the control of body weight. Notably, the effects of both leptin (Kamitakahara et al., 2018 [↗](#)) and MHFD-L (Vogt et al., 2014 [↗](#)) on targeting of AgRP projections display considerable regional specificity.

Adult mice placed on HFD display a marked hypothalamic gliosis that reveals an acute inflammatory response, which presages significant weight gain (Horvath et al., 2010; Valdearcos et al., 2017; Spencer et al., 2019; Cansell et al., 2021). This hypothalamic neuroinflammation is characterized by marked changes in the density and morphology of microglia that are most pronounced in the ARH (Thaler et al., 2012; Valdearcos et al., 2014). Microglia are the resident myeloid cells of the CNS and respond to a broad array of circulating signals, including nutrients such as saturated fats and carbohydrates (Valdearcos et al., 2014; Nadjar et al., 2017; Leyrolle et al., 2019; Butler et al., 2020). Moreover, activation of microglia alone is sufficient to stimulate food intake and promote weight gain in adult mice, and perturbations that block activation of microglia reduce the metabolic disruption associated with neuroinflammation (Valdearcos et al., 2017; Rosin & Kurrasch 2019; Sun et al., 2024).

Because of their established role as nutrient-sensing sentinels of hypothalamic neuroinflammation, and their documented participation in neural development (Stevens et al., 2007; Schafer et al., 2012; Stephan et al., 2012), microglia have been proposed as possible mediators of developmental programming caused by nutritional perturbations (Rosin & Kurrasch 2019; Folick et al., 2021).

Evidence from several of lines of investigation indicate that microglia have multiple roles in brain development (Bilbo & Schwarz 2009; Tremblay et al., 2011; Miyamoto et al., 2016; Li et al., 2019). Although microglia were initially thought to remain quiescent until activation by neuroinflammation, in vivo imaging experiments demonstrated continual activity of their cellular processes, which actively survey their local environment (Nakajima & Kohsaka 2001; Town et al., 2005; Wake et al., 2009; Stowell et al., 2018), including direct contact with axons and dendrites (Schafer et al., 2013). In addition to impacting neuronal number and initial formation of neural circuits through effects on axon targeting, microglia are thought to play an important role in synaptic refinement through selective elimination of synapses, a process termed synaptic pruning (Paolicelli et al., 2011; Schafer et al., 2012; Hong et al., 2016). Thus far, the majority of developmental microglia studies have focused on their role in cortical or hippocampal circuits. However, transcriptional profiling suggests a great deal of regional and temporal variation in microglial cell type and activity (Hammond et al., 2019; Masuda et al., 2020; Young et al., 2021), and the effects of dietary interventions have largely focused on the ARH. Here we used Iba1 immunostaining to visualize and measure microglial morphology in regions known to receive AgRP inputs. Morphological parameters of microglia were quantified in the PVH and ARH nuclei of the hypothalamus, as well as the bed nuclei of the stria terminalis (BST), a major limbic target of AgRP neurons, in offspring exposed to MHFD-L and compared with offspring that were raised on a normal chow diet (NCD). We also used genetically-targeted axonal labeling of AgRP neurons to directly visualize cellular interactions between microglia and labeled AgRP terminals in the PVH and ARH to determine if MHFD-L stimulates synaptic pruning in these regions. The results demonstrate regionally-specific changes to microglia in the PVH of MHFD-L offspring, that are temporally restricted to the period when AgRP neurons innervate the PVH. In addition, the axon labeling experiments confirm a significant decrease in AgRP innervation of the PVH in MHFD-L offspring, and for the first time provide direct evidence of microglial-mediated synaptic pruning of AgRP terminals in the hypothalamus. Microglial depletion experiments determined that the significant decrease in AgRP innervation of the PVH observed in MHFD-L offspring requires normal densities of microglia, and that microglia are required for the weight gain seen in offspring of MHFD-L dams at weaning. However, we did not detect a significant effect of MHFD-L on the degree of synaptic pruning in the PVH, suggesting an alternative microglial signaling mechanism yet to be identified.

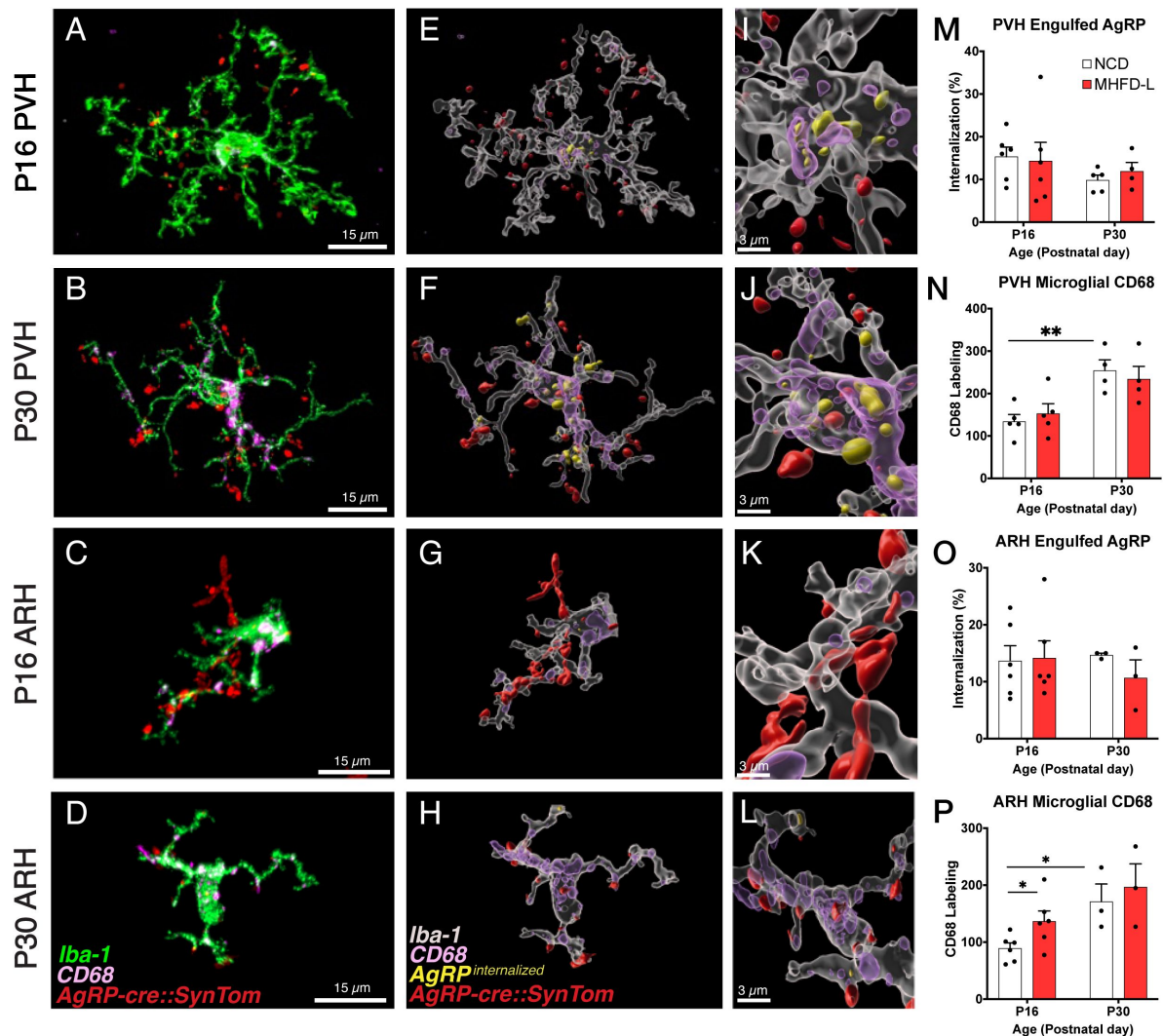
## 2 Results

### 2.1 Microglia exhibit morphological changes in the PVH in response to MHFD-L during postnatal development

To assess the impact of MHFD-L on microglia in the brains of postnatal mice, we used Iba1 immunohistochemistry and confocal microscopy to visualize the distribution and morphology of microglial cells in the PVH. Discrete regions of interest were imaged and a 3D modeling analysis pipeline was used to measure structural changes in microglia (Figure 1Ai-iii). We found that in the PVH, the overall size of microglia was significantly enhanced in offspring exposed to MHFD-L than those raised on NCD at P16 (Figure 1B,C). The increased size of microglia observed in MHFD-L offspring is demonstrated through a 44% increase in the complexity of microglial process branching, as determined by Sholl analysis (Figure 1F), as well as an 87% increase in microglial process length (Figure 1G). Additionally, the volume of microglial cells (volume measurements include cell body and processes; Figure 1I), and the spatial territory they occupy (Figure 1H), was nearly doubled in MHFD-L offspring, while the density of microglia between the dietary groups remained unaltered at P16 (Figure 1J). The changes in microglia between NCD and MHFD-L mice at P16 (Figure 1B,C) appeared to be transient, because by P30 (Figure 1D,E), we did not detect changes in microglial size (process complexity, Figure 1F; and length, Figure 1G), microglia cell territory (Figure 1H), microglial cell volume (Figure 1I), or density of microglial cells (Figure 1J) between MHFD-L and NCD offspring. The density of microglia in the PVH was reduced from P16 to P30, although this was independent of dietary treatment (Figure 1J). We also used Cre-dependent targeting of synaptophysin-tdTomato to axons of AgRP neurons to assess the density of AgRP terminals in PVH regions of interest. The results confirmed that AgRP terminal density is significantly lower in the brains of MHFD-L offspring compared with NCD controls at both P16 and P30 (Figure 1K). Taken together, these results suggest that exposure to MHFD-L causes changes to the morphology of microglia that are consistent with enhanced activity and surveillance of their immediate microenvironment. Furthermore, the effects of MHFD-L on microglial morphology in the PVH correspond to an accompanying decrease in the density of AgRP inputs to the PVH.

### 2.2 Microglia do not exhibit morphological changes in the ARH or BST in response to MHFD-L exposure during postnatal development

The ARH houses the cell bodies of AgRP neurons, and their number is established primarily during prenatal development (Ishii & Bouret 2012). We evaluated microglia morphology in the ARH of postnatal mice by using the same 3D modeling pipeline shown in Figure 1. In contrast to our findings in the PVH, microglial size and density in the ARH were not significantly different in NCD and MHFD-L offspring at P16 (Figure 2A,B; quantified in 2E-I). By P30 (Figure 2C,D) there was a 67% increase in process length (Figure 2F) of NCD mice compared to their P16 counterparts, but no significant differences in microglial morphology in the ARH were detected between the dietary treatment groups (Figure 2A-G). The volume of individual microglial cells (Figure 2H), as well as the spatial territory they occupy (Figure 2G), increased from P16 to P30, while the total number of microglial cells was reduced by nearly half (Figure 2I). We also measured numbers of AgRP neuronal cell bodies in the ARH in brains derived from NCD and MHFD-L offspring and confirmed that the number of AgRP neurons in the ARH is also resistant to MHFD-L exposure at both P16 and P30 (Figure 2J). In addition, we evaluated microglia and AgRP terminals in the anterolateral part of the BST (Figure 3), an extrahypothalamic target of AgRP neurons innervated during the lactational period (Bouret et al., 2004a; Cansell et al., 2012; Barbier et al., 2021). As was found for the ARH, neither the size nor number of



**Figure 1.**

### MHFD-L: Microglial morphology in the PVH.

(A) Image analysis pipeline. Fluorescence image of an Iba1-immunostained microglial cell in the PVH (Ai). Confocal images through labeled cells were used to generate 3D reconstructions (Aii), which were then used to create 3D models of microglial cells by using the Filaments tool in Imaris. Polyhedrons were generated around each cell using the Convex Hull function of Imaris to estimate the total tissue “territory” occupied by the microglial cell (Aiii). (B-E) Images of microglial cells (green) and labeled AgRP terminals (red) in the PVH of mice at P16 (B,C) or P30 (D,E) that were raised on NCD (B, D) or MHFD-L (C, E). Graphical comparisons between groups to show that MHFD-L increased microglial ramification complexity (F), process length (G), microglial cell territory (H), and cell volume (I) at P16. The density of microglia in the PVH decreased between P16 and P30, irrespective of diet (J). Density of AgRP terminals were decreased in MHFD-L offspring at both P16 and P30 (K). Bars represent the mean  $\pm$  SEM and each point represents one animal. \* $P < .05$ , \*\* $P < .005$ . Abbreviations: MHFD-L, maternal HFD during lactation; NCD, Normal Chow Diet; PVH, paraventricular nucleus of the hypothalamus.



microglia in the BST were significantly different between NCD and MHFD-L offspring at P16 or P30 (**Figure 3A-D** [↗](#); quantified in 3E-I). Similarly, the density of AgRP terminals in the same region of interest was not affected by MHFD-L exposure (**Figure 3J** [↗](#)), although by P30 the number of AgRP terminals in the BST increased by 61% compared to their P16 counterparts (**Figure 3J** [↗](#)). Taken together, these data suggest that microglia in the ARH and BST are resistant to the morphological changes that occur in the PVH of MHFD-L offspring, suggesting a notable degree of spatial specificity in the role of hypothalamic microglia during postnatal development.

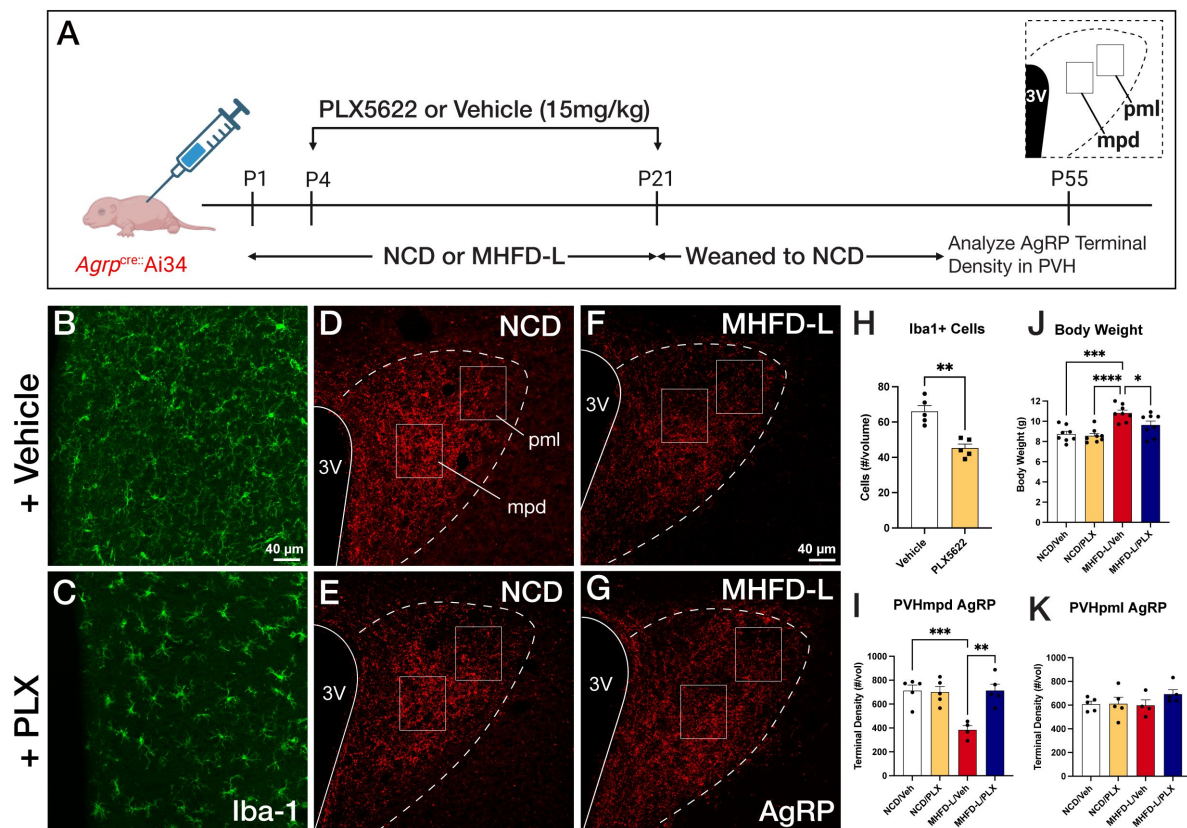
## 2.3 Microglia are required for changes in AgRP terminal density in PVH and body weight associated with MHFD-L exposure

To determine if microglia are required for the observed changes in AgRP inputs to the PVH of MHFD-L offspring, the colony stimulating factor 1 receptor (CSF1R) inhibitor PLX5622 was administered from P4-P21 via daily intraperitoneal (i.p.) injection (**Figure 4A** [↗](#)). These postnatal treatments resulted in a significant decrease in microglia detected in the hypothalamus at P55 relative to age-matched controls (**Figure 4B,C** [↗](#); quantified in 4H). Notably, the PLX5622 treatments blocked the reduction in AgRP fiber density observed in the medial dorsal parvocellular part of the PVH (PVHmpd) of vehicle-treated MHFD-L offspring to a level that was comparable to that of NCD offspring (**Figure 4D-G** [↗](#); quantified in 4I). In contrast, MHFD-L exposure did not affect the density of AgRP fibers in the lateral posterior magnocellular compartment of the PVH (PVHmpl) and no significant difference in AgRP fiber density was detected between NCD and MHFD-L offspring treated with either PLX5622 or vehicle (**Figure 4D-G** [↗](#); quantified in 4K), suggesting target specificity for the microglial-mediated effects on development of AgRP inputs to the PVHmpd.

Depletion of microglia also appeared to protect against the increase in body weight normally observed in MHFD-L mice. In keeping with previously published results (Vogt et al., 2014 [↗](#)), the body weight of vehicle-treated MHFD-L animals was significantly greater (24%), compared with that of NCD animals at weaning (**Figure 4J** [↗](#)). However, MHFD-L mice treated with PLX5622 during lactation exhibited significantly lower weights at weaning compared to those of vehicle-treated MHFD-L animals (**Figure 4J** [↗](#)). Taken together, these findings suggest that microglia mediate target-specific effects of MHFD-L exposure on the innervation of the PVH by AgRP neurons, and that microglia play a role in mediating the effects of MHFD-L on body weight.

## 2.4 Engulfment of AgRP terminals by microglia in the PVH and the ARH

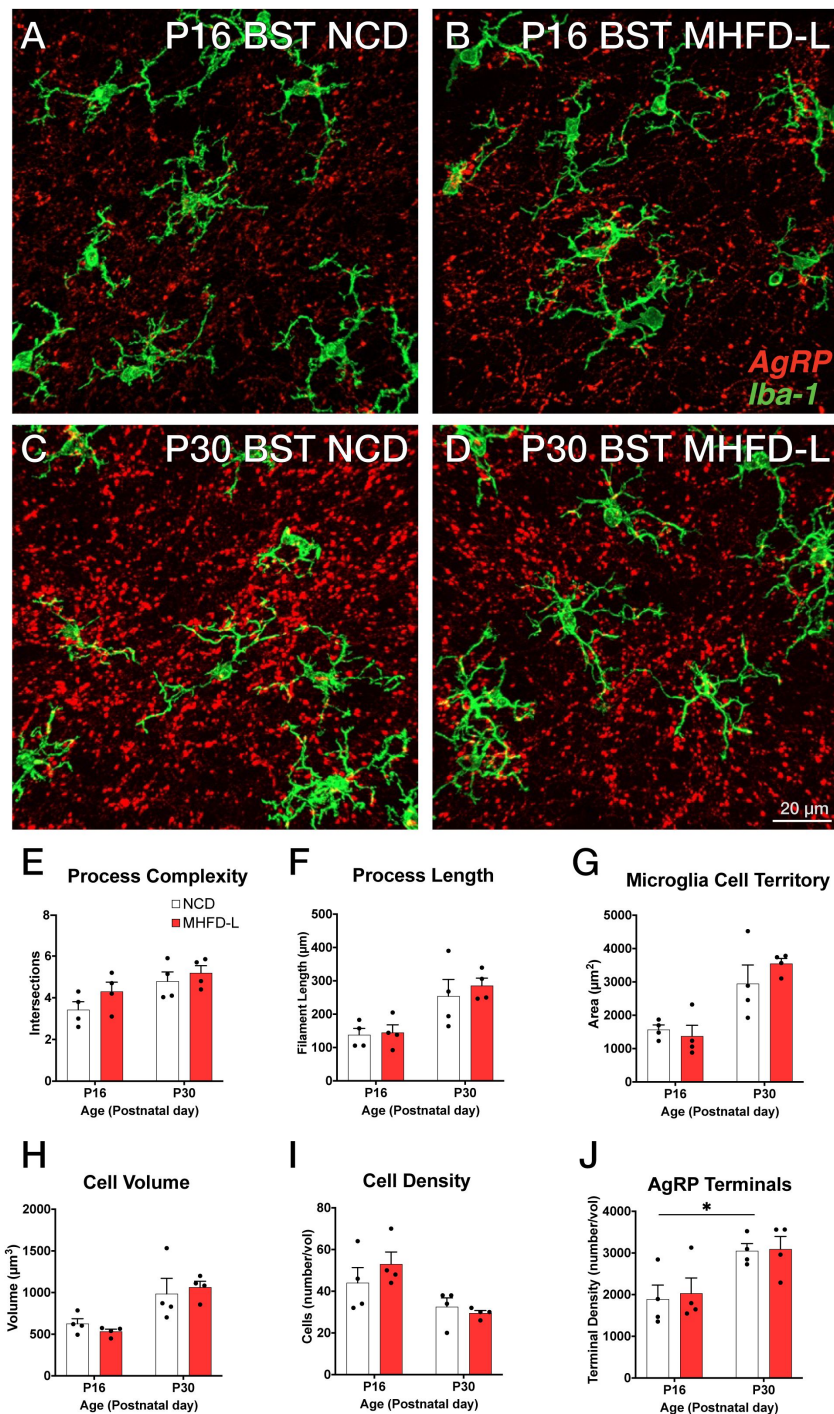
Microglia are thought to impact development of neuronal connections through an active engulfment mechanism and the lysosomal associated membrane protein CD68 has been implicated in this process. Here, we used immunohistochemistry to visualize the presence of in Iba1-labeled microglia in mice with genetically targeted labeling of AgRP terminals (**Figure 5** [↗](#)). Many apparent contacts between microglial processes and AgRP terminals in the PVH and ARH were observed at P16 and P30, including internalized AgRP terminals (**Figure 5A-L** [↗](#)). However, the extent of internalization did not appear to be influenced by MHFD-L exposure; there were no significant differences between internalized AgRP terminals in MHFD-L and NCD offspring at P16, in either the PVH or ARH (**Figure 5M,O** [↗](#)). Similarly, we did not detect a statistically significant difference in microglial CD68 levels in the PVH between diet groups at either P16 or P30 (**Figure 5N** [↗](#)). Consistent with previous reports (Wong et al., 2005 [↗](#); Hart et al., 2012 [↗](#)), the density of CD68 labeled profiles nearly doubled in the PVH between P16 and P30 (**Figure 5N** [↗](#)), as microglia become more phagocytic with age. In the ARH, CD68 staining also increased between P16 and P30 (**Figure 5P** [↗](#)), supporting the notion that microglia increase their phagocytic capacity with age.



**Figure 2.**

### MHFD-L: Microglial morphology in the ARH.

Microglial cells (green) and labeled AgRP terminals (red) in the ARH of mice at P16 (A, B) or P30 (C, D) that were exposed to NCD (A, C) or MHFD-L (B, D). Graphical comparisons between groups to show that microglial ramification complexity (E) remained the same, regardless of age or diet. Microglial process length (F), cell territory (G), and cell volume (H) increased between P16 and P30, but were not changed as a result of diet. The density of microglia in the ARH decreased between P16 and P30, irrespective of diet (I). There were no apparent changes in numbers of AgRP neurons (J). Bars represent the mean  $\pm$  SEM and each point represents one animal. \* $P < .05$ , \*\* $P < .005$ . Abbreviations: ARH, arcuate nucleus of the hypothalamus; MHFD-L, maternal HFD during lactation; NCD, Normal Chow Diet.

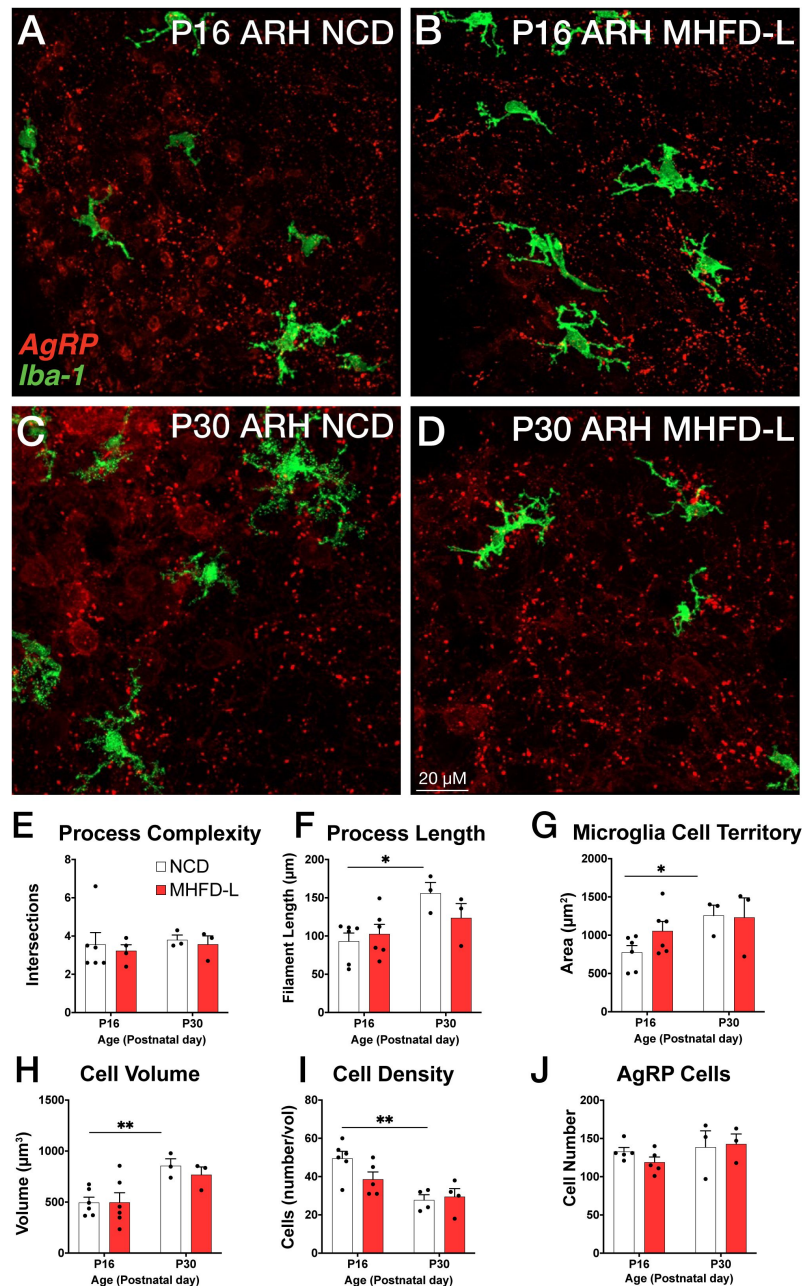


**Figure 3.**

### MHFD-L: Microglial morphology in the BST.

Microglial cells (green) and labeled AgRP terminals (red) in the BST of mice at P16 (A, B) or P30 (C, D) that were exposed to NCD (A, C) or MHFD-L (B, D). Graphical comparisons between groups to show that microglial ramification complexity (E), process length (F), cell territory (G), and cell volume (H) did not significantly change between P16 and P30, nor were they changed as a result of diet. The density of microglia in the BST decreased between P16 and P30, irrespective of diet (I). The density of AgRP terminals increased between P16 and P30, but there was no effect of maternal diet (J). Bars represent the mean  $\pm$  SEM and each point represents one animal. \* $P < .05$ . Abbreviations: BST, bed nucleus of the stria terminalis; MHFD-L, maternal HFD during lactation; NCD, Normal Chow Diet.





**Figure 4.**

### Microglial depletion during lactation period.

Schematic of MHFD-L exposure and PLX5622 treatment experimental design (A). Microglial cells (green) in the PVH of adult mice (P55) treated during lactation with vehicle (B) or PLX5622 (C). Images of labeled AgRP terminals (red) to illustrate the density of labeling in distinct compartments of the PVH (white boxes denote locations of ROIs) of NCD offspring (D,E) and MHFD-L offspring (F,G). Graphical comparisons to illustrate the effects of postnatal PLX5622 treatments on microglia density in the PVH (H), body weight (J) and the density of AgRP terminals in the PVHmpd (I) and PVHpml (J). Bars represent the mean  $\pm$  SEM and each point represents one animal. Unpaired t-test was used to compare cell number in 4H; two-way ANOVA was used to test for differences in group means, followed by Tukey's multiple comparisons posthoc test to identify specific group differences in 4I-K. P-values less than 0.05 were considered significant; \* $P < .05$ , \*\* $P < .005$ , \*\*\* $P < .0005$ . Abbreviations: AgRP, agouti-related peptide; CSF1R, Colony Stimulating Factor 1 Receptor; MHFD-L, maternal HFD during lactation; PVH, paraventricular nucleus of the hypothalamus; mpd, medial parvocellular compartment of the PVH; pml, posterior magnocellular compartment of the PVH.

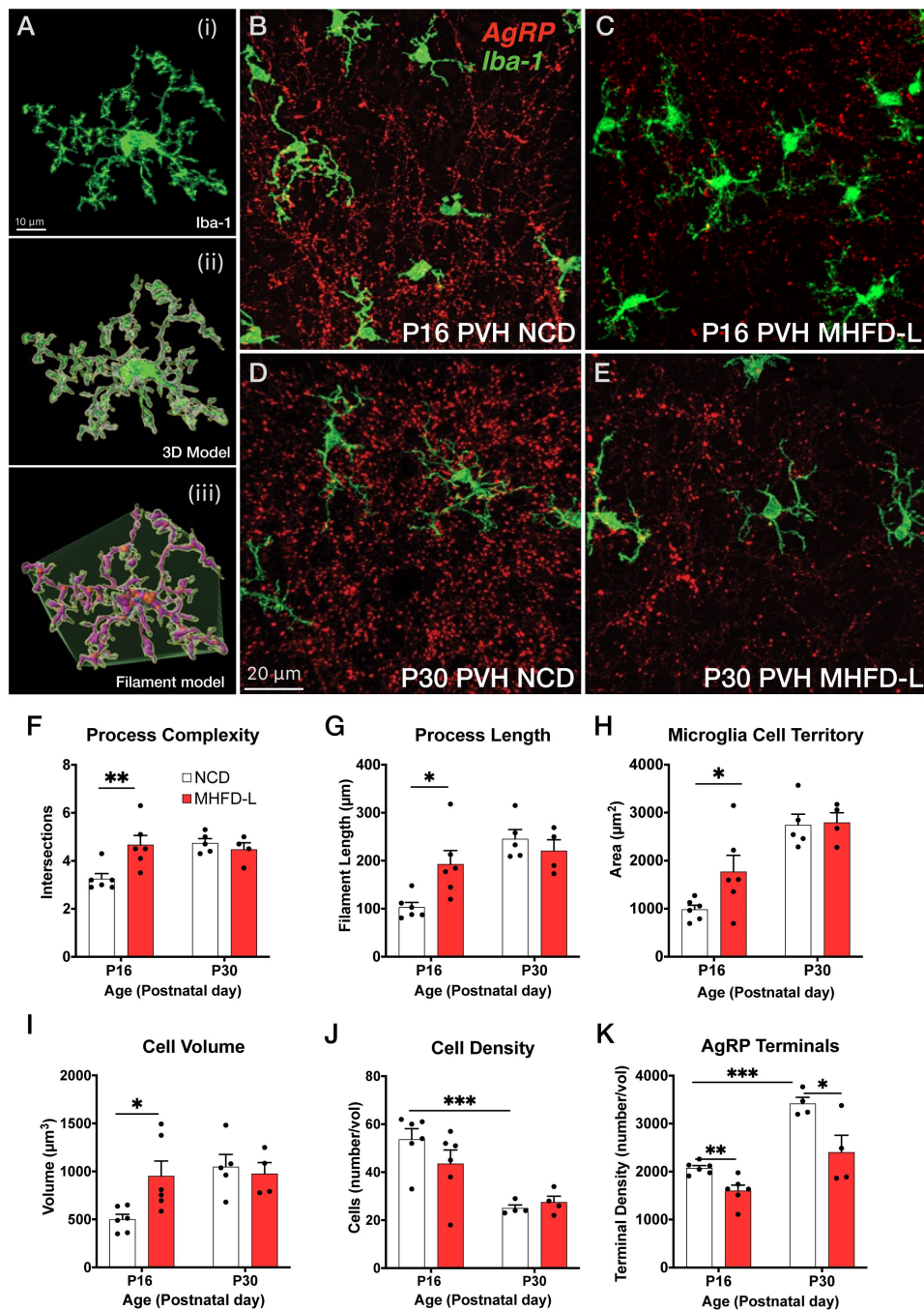
Nevertheless, our analysis demonstrates that microglia interact directly with AgRP terminals, with clear evidence of engulfment. MHFD-L exposure does not appear to promote microglia-mediated engulfment, at least not in the specific PVH and ARH domains examined.

## 3 Discussion

It is well established that microglia are responsive to HFD exposure in adult rodents (Thaler et al., 2012 [↗](#); Morari et al., 2014 [↗](#); Valdearcos et al., 2014 [↗](#); Baufeld et al., 2016 [↗](#); Valdearcos et al., 2017 [↗](#)) and multiple lines of evidence support an important role for microglia in mediating key aspects of neural circuit development (Checchin et al., 2006 [↗](#); Hoshiko et al., 2012 [↗](#); Li et al., 2012 [↗](#); Hagemeyer et al., 2017 [↗](#)). Here we demonstrate that microglia are required for significant elevations in body weight that emerge from postnatal exposure to HFD and are associated with a sustained decrease in the density of afferents from AgRP neurons to the PVH in offspring. Exposure to MHFD-L caused distinct morphological changes to microglia that may be consistent with enhanced activity, which were observed in the PVH, but not the ARH or BST. Moreover, the morphological changes to microglia observed appear to be primarily limited to the critical period for development of AgRP inputs to PVH neurons in MHFD-L offspring. Although our results demonstrate that microglia engage in engulfment of AgRP terminals in the PVH during development, synaptic pruning by microglia does not appear to represent the cellular mechanism mediating the effects of MHFD-L exposure on innervation of the PVH by AgRP neurons.

### 3.1 A Role for Microglia in Mediating Body Weight Changes Observed in MHFD-L Offspring

Maternal HFD exposure during perinatal development leads to increased body weight, fat content, and susceptibility to diet-induced obesity in offspring (Samuelsson et al., 2008 [↗](#); Tamashiro et al., 2009 [↗](#); Masuyama & Hiramatsu 2014 [↗](#)). The postnatal period, which corresponds to lactation, and when mice derive their nutrition primarily from milk, is especially sensitive to nutritional environment and exposure to HFD that is exclusive to this period plays a particularly dominant role in specifying metabolic phenotype later in life (Chen et al., 2008 [↗](#); Sun et al., 2012; Vogt et al., 2014 [↗](#)). In the present study, global depletion of microglia with the CSF1R inhibitor PLX5622 blocked the ability of MHFD-L to increase body weight in mice by weaning at P21, indicating microglia may mediate metabolic changes caused by MHFD-L exposure. Consistent with this, microglial depletion with PLX5622 in adult mice mitigates the effects of HFD exposure (Rosin & Kurrasch 2019 [↗](#)), and our results indicate that microglia may function similarly in offspring during postnatal life to effect changes in body weight, even if the maternal HFD exposure is restricted to the lactation period. Further studies are required to define the long-term metabolic profile resulting from developmental microglial manipulations. However, given the abundant literature on sustained impact of MHFD-L on metabolic phenotype, enduring disruptions are likely. It should be noted that PLX5622 treatment is not spatially limited to the PVH or ARH, leaving open the possibility that the effects of microglial depletion on body weight occur outside of these nuclei, or are due to collective activation of microglia in multiple components of feeding circuitry (Green et al., 2020 [↗](#)). Localization of the specific site of action for microglial specification of mature body weight during development will require utilization of specific markers for hypothalamic microglia that account for regional and phenotypic heterogeneity, perhaps through intersectional genetic methods and combinatorial pharmacology (Hammond et al., 2019 [↗](#); Kim et al., 2021 [↗](#)).



**Figure 5.**

### Microglial interaction with AgRP axon terminals in the PVH and ARH.

(A-D) Representative images of microglial cells (green), labeled AgRP terminals (red), and CD68 (lysosomal associated membrane protein and phagocytic capacity marker, pink) that compare their cellular relationships in the PVH (A,B) and ARH (C,D) at P16 and P30. (E-H) Digital 3D reconstructions of cells shown in (A-D) after application of filaments tool to visualize internalized AgRP terminals. (I-L) Cells shown in E-H after application of digital zoom to more clearly illustrate engulfment of labeled AgRP terminals by microglia and location of CD-68 labeled profiles. (M-P) Graphical comparisons between groups to illustrate the effects of age and MHFD-L exposure on CD68 expression and AgRP terminal engulfment. Bars represent the mean  $\pm$  SEM and each point represents one animal. \* $P < .05$ , \*\* $P < .005$ . Abbreviations: AgRP, agouti-related peptide; ARH, arcuate nucleus or the hypothalamus; CD68, Cluster of Differentiation 68; MHFD-L, maternal HFD during lactation; PVH, paraventricular nucleus of the hypothalamus.

### 3.2 MHFD-L Induces Spatially Limited Changes in Microglial Morphology

Morphological changes in microglia have been reported in response to a variety of environmental exposures. In the hypothalamus, adult mice fed a HFD show both proliferation and changes in microglial process length and complexity (Thaler et al., 2012 [↗](#); Valdearcos et al., 2017 [↗](#)). In our studies, MHFD-L exposure caused a marked increase in the overall size of microglia in the PVH that is related to increases in both the length and branching complexity of immunolabeled Iba1 cellular processes. This increase in the territory occupied by microglia in the PVH was not accompanied by an increase in microglial number, nor were numbers of microglia affected in the PVH by MHFD-L exposure. However, in contrast to the PVH, changes in microglial morphology were not observed in the ARH in response to MHFD-L, although we did observe an overall increase in process length in the ARH between P16 and P30 of both MHFD-L and NCD mice. This finding is consistent with previously published reports on microglial maturation (Sun et al., 2024 [↗](#)). The spatially restricted enhancement of microglial activation in the PVH resulting from MHFD-L exposure appears to contribute to an expansion of parenchymal territory surveilled by PVH microglia, as reflected in the volume measurements accomplished with geometrical modeling of process length and complexity. This interpretation is supported by *in vitro* and *in vivo* observations of enhanced process extension and increased neuronal interactions resulting from inflammatory activation of microglia (Wake et al., 2009 [↗](#); Schafer et al., 2013 [↗](#); Dissing-Olesen et al., 2014 [↗](#); Stowell et al., 2018 [↗](#); Bolton et al., 2022 [↗](#)).

As in the ARH, we did not observe comparable changes in microglial morphology in the BST, an extrahypothalamic target of AgRP neurons innervated during the lactation period (Cansell et al., 2012 [↗](#); Barbier et al., 2021 [↗](#)). These observations underscore the remarkable molecular heterogeneity of microglial phenotypes that appear to occupy various hypothalamic niches during development, and may have equally diverse developmental roles and responses to environmental signals (Bilbo & Schwarz, 2012 [↗](#); Frost & Schafer, 2016 [↗](#); Li & Barres, 2018 [↗](#); Ngozi & Bolton, 2022 [↗](#)). Moreover, the observed morphological changes in the PVH caused by MHFD-L appear to be transient as there no significant differences in microglial processes by P30, and the density of microglial cells in the PVH was significantly reduced in the older mice, suggesting a decline in overall activity. That the observed changes in microglial morphology occur within the critical period for development of AgRP projections to the PVH suggests a possible role for microglia linking nutrition with specification of axonal targeting (Kamitakahara et al., 2018 [↗](#)).

### 3.3 Microglia Mediate Impaired Innervation of the PVH by AgRP Neurons

AgRP neuronal projections develop primarily during the first two weeks of life, which corresponds to a critical period for the neurotrophic action of leptin on axonal outgrowth and targeting of AgRP inputs to distinct components of the PVH (Bouret et al., 2004a [↗](#); Elson & Simerly, 2015 [↗](#)). Exposure to MHFD-L during this critical developmental window permanently impairs AgRP projections to the PVH, DMH and LH (Vogt et al., 2014 [↗](#)) that are associated with increased body weight in adulthood. Here we confirm that MHFD-L impairs innervation of the PVH, and this defect was apparent by P16. The PVH is innervated by AgRP neurons between P8 and P10 (Bouret et al., 2004a [↗](#)). We found that depletion of microglia with PLX5622 between P4 and P21, a period that aligns not only with AgRP innervation of the PVH but also with maximum changes in microglial morphology, blocked the effects of MHFD-L on AgRP terminals in the PVH. However, this partial rescue of innervation appeared to be limited to the PVHmpd, suggesting that there is a regional specialization in the activity of microglia in the PVH. We did not observe a change in AgRP innervation of the BST in MHFD-L offspring that mirrored the changes seen in the PVH of the same animals, further supporting the conclusion that microglia display at least some degree of spatial heterogeneity in mediating site-specific alterations in AgRP axon targeting.



Microglia are known to impact a variety of developmental events, including alterations in cell number through programmed cell death or neurogenesis, as well as axonal targeting and remodeling of neural circuits (Frost & Schafer, 2016 [↗](#); Li & Barres 2018 [↗](#)). Because PLX5622 causes a decrease in microglia in the ARH as well as in the PVH, our microglial depletion studies do not eliminate the possibility that microglia may act locally on AgRP neurons to affect axonal growth. It is unlikely that the impaired innervation of the PVH observed in MHFD-L offspring is due to a reduction in the number of AgRP neurons in the ARH (Vogt et al., 2014 [↗](#); Valdearcos et al., 2017 [↗](#)). Sun and colleagues (2024) [↗](#) reported that microglial depletion during postnatal life increases numbers of NPY neurons, as well as enhances local densities of AgRP fibers in the ARH, possibly through enhanced formation of perineuronal nets. Depletion of microglia during gestation causes a significant decrease in the number of proopiomelanocortin (POMC) neurons in the ARH and leads to an acceleration of weight gain (Rosin & Kurrasch, 2019 [↗](#)), consistent with neurogenesis of ARH neurons occurring in mid gestation and increased susceptibility to nutritional impacts during embryonic life (Ishii & Bouret, 2012 [↗](#); Elson & Simerly, 2015 [↗](#)). Interestingly, genetic deletion of leptin receptors from myeloid cells reduced numbers of POMC neurons in the ARH, suppressed POMC innervation of the PVH, and decreased microglial process complexity (Gao et al., 2017 [↗](#)). Taken together, these results suggest that leptin signaling in microglia may act at the level of the ARH to promote outgrowth of AgRP projections to the PVH, while MHFD-L activates microglia in the PVH to specify patterns of AgRP afferents that are not only regionally specific, but also target discrete subdomains of the PVH. However, whether microglia inhibit synaptogenesis or are involved in synaptic refinement through an alternative regressive mechanism will require further investigation.

### 3.4 Microglia in PVH Participate in Synaptic Pruning

Microglia have been proposed as mediators of synaptic pruning, a process whereby synapses that form early in development are eliminated as others are strengthened and maintained (Katz & Shatz, 1996 [↗](#); Sanes & Lichtman, 1999 [↗](#); Frost & Schafer 2016 [↗](#); Li & Barres, 2018 [↗](#)). Although this process has been studied most extensively in somatosensory cortex (Miyamoto et al., 2016 [↗](#)), hippocampus (Paolicelli et al., 2011 [↗](#); Wang et al., 2020 [↗](#)) and the visual system (Tremblay et al., 2010 [↗](#); Schafer et al., 2012 [↗](#)), there is evidence for involvement of microglia in synaptic pruning of immunolabeled glutamatergic terminals associated with corticotropin releasing hormone neurons in the PVH (Bolton et al., 2022 [↗](#)). In the present study, we used genetically targeted axonal labeling to provide evidence that microglia in the PVH participate in synaptic pruning of AgRP synapses during the critical period for PVH innervation, and when PVH microglia exhibit high levels of process extension. The lysosomal marker CD68 was colocalized with internalized AgRP terminals in PVH microglia, and although elevated at P30, there were no differences between offspring of MHFD-L and NCD dams. Additionally, we did not find a significant difference in the density of engulfed AgRP terminals in the PVH of MHFD-L offspring at either P16 or P30. However, enhanced engulfment of AgRP terminals in MHFD-L offspring may occur at a later point in development not assessed in this study. It is also possible that the synaptophysin-tdTomato axonal label may have been lost from pruned synapses during engulfment. A more probable interpretation is that PVH microglia participate in synaptic refinement through other cellular mechanisms, including microglial release of secreted factors such as the interleukin IL-6 (Kim et al., 2024 [↗](#)) or microglial-derived BDNF (Parkhurst et al., 2013 [↗](#)). Future studies that include cell type-specific manipulations of microglial signaling and live cell imaging may clarify these potential developmental mechanisms.

## 4 Conclusions

MHFD-L causes elevated levels of saturated carbohydrates and fats in milk (Gorski et al., 2006 [↗](#); Vogt et al., 2014 [↗](#); Calvo-Lerma et al., 2022 [↗](#)). The resulting overnutrition resulting from exposure to this enhanced diet is thought to underlie the propensity towards obesity observed in

offspring later in life (Skowronski et al., 2023 [DOI](#)). Although microglia are likely mediators of multiple neurobiological events influencing how hypothalamic circuits function during regulation of energy balance, the precise signaling mechanisms remain ill defined. There may be common molecular mechanisms underlying the effects of HFD exposure on microglial activation in adults and those occurring during postnatal development, but how these signaling events exert a lasting impact on the organization and function of feeding circuitry has not been defined. The results presented here demonstrate an important role for microglia on sculpting the density of inputs from AgRP neurons to the PVH that is not only spatially restricted, but also aligned temporally with synaptogenesis in the PVH. Furthermore, PVH microglia clearly interact directly with AgRP afferent axons during this critical period and may be refined through engulfment by microglia. However, synaptic pruning through engulfment does not appear to be sufficient to affect the significant reduction in AgRP innervation of PVH neurons observed following MHFD-L exposure, suggesting involvement of additional microglial signaling mechanisms that are not only important for specifying patterns of innervation of the PVH by AgRP neurons, but may also contribute more broadly to developmental programming of metabolic phenotype.

## 5 Materials and Methods

### 5.1 Key Resources Table

#### 5.2 Animals

All animal care and experimental procedures were performed in accordance with the guidelines of the National Institutes of Health and the Institutional Care and Use Committee of Vanderbilt University. Mice were housed at 22°C on a 12:12 h light:dark cycle provided ad libitum access to NCD (PicoLab Rodent Diet 20 #5053) and water unless otherwise specified. AgRP-Cre mice (*AgRP<sup>tm1(cre)Low</sup>/J*; stock number: 01289) and mice expressing the Cre-dependent fluorescent reporter synaptophysin-tdTomato (RCL-Syp/tdT)-D (B6;129S-*Gt(ROSA)26Sor<sup>tm34.1(CAG-Syp/tdTomato)/Hze</sup>/J*; stock number: 012570) were obtained from the Jackson Laboratory (Bar Harbor, ME) and maintained in our colony at Vanderbilt University. To visualize AgRP inputs, *AgRP*-Cre mice were crossed with Ai34D mice to generate *AgRP*-Cre::Ai34D mice, as described previously (Biddinger et al., 2020 [DOI](#)).

To generate offspring of dams exposed to HFD during lactation (MHFD-L), mice had ad libitum access to NCD (PicoLab Rodent Diet 20 #5053: 25% protein; 62% carbohydrates; 13% fat; 4 kcal/g energy density) prior to and during mating. On the first postnatal day (P1) all litters were adjusted to 7 pups to normalize nutrition and dams were switched to either HFD (Research Diets D12451: 20% protein; 35% carbohydrate; 45% fat; 4.7 kcal/g energy density) or kept on the same NCD. The dams remained on either HFD or NCD throughout lactation, and offspring were weaned onto the same normal chow diet that the dams had received prior to MHFD-L treatment, regardless of lactation dietary condition, and remained on NCD until they were processed for perfusion.

#### 5.3 PLX5622 Microglia Depletion

To reduce microglia during postnatal development, mouse pups were treated daily from P4 to P21 with either PLX5622, a colony stimulating factor 1 receptor (CSF1R) inhibitor or DMSO vehicle via intraperitoneal injection (Riquier & Sollars 2020 [DOI](#)). Briefly, PLX5622 hemifumarate solid (Cat. #HY114153A MedChemExpress, Monmouth Junction, NJ, USA) was suspended in DMSO at a concentration of 172 mg/ml. The injection working solution was prepared to include 20% Kolliphor RH40 diluted in PBS, which resulted in doses with a 6.5mg/ml PLX5622 concentration and injection concentration of 15 mg/kg.

| Reagent type or resource                  | Designation                                                                              | Source or reference     | Identifiers                              | Additional information |
|-------------------------------------------|------------------------------------------------------------------------------------------|-------------------------|------------------------------------------|------------------------|
| Genetic reagent<br>( <i>M. musculus</i> ) | <i>AgRP<sup>tm1(cre)Low</sup>/J</i>                                                      | Jackson Laboratory      | Stock #: 012899<br>RRID: IMSR_JAX:012899 | MGI ID: J:140858       |
| Genetic reagent<br>( <i>M. musculus</i> ) | Ai34(RCL-Syp/tdT)-D<br>(B6;129S-Gt(ROSA)26Sor <sup>tm34.1(CAG-Syp/tdTomato)/Hze</sup> /J | Jackson Laboratory      | Stock #: 012570<br>RRID: IMSR_JAX:012570 | MGI ID: J:170755       |
| Antibody                                  | Rabbit polyclonal anti-Iba1                                                              | FUJIFILM Wako           | Cat. #: 019-19741<br>RRID: AB_839504     | IHC (1:2000)           |
| Antibody                                  | Rat monoclonal anti-CD68 [FA-11]                                                         | Abcam                   | Cat. #: ab53444<br>RRID: AB_869007       | IHC (1:500)            |
| Antibody                                  | Donkey polyclonal anti-rabbit Alexa Fluor 488                                            | ThermoFisher Scientific | Cat. #: A32790<br>RRID: AB_2762833       | IHC (1:500)            |
| Antibody                                  | Donkey polyclonal anti-rat Alexa Fluor 647                                               | ThermoFisher Scientific | Cat. #: A48272<br>AB_2893138             | IHC (1:500)            |
| Chemical compound                         | PLX5622 hemifumarate, CSF1R inhibitor                                                    | MedChemExpress          | Cat. #: HY114153A                        |                        |
| Software, Algorithm                       | Imaris                                                                                   | Bitplane                | V9.5                                     |                        |
| Software, Algorithm                       | GraphPad Prism                                                                           | Prism                   | Prism 10                                 |                        |

## Key Resources Table

## 5.4 Immunohistochemistry

Mice were perfused at P16 and P30 and processed for immunofluorescence by using primary antibodies to Iba1 (1:2000; FUJIFILM Wako, Osaka, Japan) to visualize microglia and CD68 (1:500; Abcam, Cambridge, MA, USA) to assess phagocytic capacity of the microglia. Mice were first anesthetized with tribromoethanol (TBE) and then perfused transcardially with cold 0.9% saline, followed by cold fixative (4% paraformaldehyde in borate buffer, pH 9.5) for 10 minutes. After fixative perfusion, brains were removed from the skull and postfixed in the same fixative overnight. The next day, the tissue was transferred to 20% sucrose for cryoprotection overnight. A freezing-stage sliding microtome was used to collect 30  $\mu\text{m}$ -thick coronal sections and free-floating tissue sections were stored in cryoprotectant solution at  $-20^{\circ}\text{C}$  until further processing. To prepare tissue for immunohistochemical processing, brain sections were removed from cryoprotectant and rinsed several times in 0.02M KPBS. Free-floating sections were incubated in blocking buffer containing 2% normal donkey serum and 0.3% Triton-X 100 in 0.02M KPBS overnight at  $4^{\circ}\text{C}$ . Tissue sections were incubated in the same blocking buffer with primary antibodies for 48 hr at  $4^{\circ}\text{C}$ . Following primary antibody incubation, sections were rinsed several times in KPBS, and then incubated in the appropriate species-specific fluorophore-conjugated Alexa-Fluor secondary antibodies for 1 hr at RT. Tissue sections were again rinsed several times with KPBS, and mounted onto charged microscope slides and coverslipped using ProLong antifade mounting medium (Life Technologies, Carlsbad, CA, USA).

## 5.5 Image Acquisition and Analysis

Sections through the PVH, ARH, and BST were identified and morphological features of the nuclei were visualized with Hoescht 33342, which were then used to define matching regions of interest (ROI) for quantitative analysis carried out by a user blind to treatment group. Confocal image stacks were collected using a laser scanning confocal microscope (Zeiss LSM 800) for each ROI through the entire thickness of the region at a frequency of  $0.1\mu\text{m}$  using the 40x objective. Imaris visualization software (Bitplane V9.5, Salisbury Cove, ME, USA) was used to create 3D reconstructions of each multichannel set of images.

Profiles of Iba1-immunolabeled microglia were segmented and skeletonized by using the Filaments tool in Imaris to quantify changes in microglia structure. Sholl analysis (Norris et al., 2014 [DOI](#)), was performed on the skeletonized structures to determine the branching complexity of microglial processes. Briefly, 3D concentric spheres are drawn around each selected microglial cell and contact points between microglial processes and the spheres counted. 3D reconstructions of the microglia were also evaluated for process length and cell volume. To estimate the 3D space occupied by each microglial cell, a polyhedron was drawn around the microglia by using the built-in Convex Hull function under Filaments to estimate the regional volume occupied by each cell analyzed.

In order to assess cellular interactions between AgRP terminals and microglia, 3D renderings of microglia and AgRP terminals were used to determine densities of contact points between AgRP terminals and microglial processes by using an Imaris MATLAB script to automate the analysis. AgRP terminals were reconstructed as “spots” of 0.8  $\mu\text{m}$  diameter (corresponding to the largest measured size) and their total number was calculated in each ROI. Briefly, the automatic detection algorithm applies a 3D Mexican hat filter using the spot size and then locates the spot centroid at the local maxima of the filtered image. The number of spots located at no more than  $1\mu\text{m}$  from the microglia surface was automatically determined as an estimate of contact points between AgRP terminals and microglia. Next, spots that were determined to be more than  $0.5\mu\text{m}$  away from internal microglial surfaces were determined to be internalized by the microglia and counted as engulfed AgRP terminals. Intracellular CD68 levels in microglia were estimated by segmentation of CD68 labeling profiles in image stacks followed by creation of 3D renderings and their volume



computed. Finally, to assess the effect of MHFD-L on numbers of AgRP neurons in the ARH, AgRP neuronal cell bodies were identified by aligning synaptophysin-tdTomato labeled somal profiles with Hoescht 33342-stained nuclei and counting the number of visualized neurons manually.

## 5.6 Statistical Analyses

Data are presented as group mean values  $\pm$  SEM, as well as individual data points. Statistical analyses were performed using GraphPad Prism software (Version 10). Unpaired t-tests were used to compare data between two groups. In the microglia depletion experiment, two-way analysis of variance (ANOVA) was used to test for differences in group means, followed by Tukey's multiple comparisons post hoc test to identify specific group differences. Differences between groups were considered statistically significant at  $p < 0.05$ .

## Data Availability

All data collected in this study are included in the manuscript and have been deposited in the Open Science Framework.

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## Additional information

### Author Contributions

HNM-R and RBS conceived of this research and designed experiments. HNM-R conducted experiments with technical support from JEB and MNB. HNM-R and JEB analyzed data and prepared figures. HNM-R and RBS wrote the manuscript and all authors participated in editing the manuscript.

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**Reviewer #1 (Public review):**

## Summary:

This paper shows that maternal high-fat diet during lactation changes microglia morphology in the PVN, potentially to acquire a more active state. Further, the authors reveal that PVN microglia engulf AgRP terminals in the PVN during postnatal development, a previously unrecognized behavior. A notable finding of this paper is that pharmacological reduction of microglial cells can reverse weight gain and terminal loss in the offspring under maternal high fat diet conditions, even though an increase in microglial engulfment of AgRP+ terminals was not observed, suggesting an alternative mechanism. The data support these findings, although questions remain regarding the efficacy and timing of the pharmacological microglial knockdown.

## Strengths

(1) The impact of microglia on hypothalamic synaptic pruning is poorly characterized, and thus, the findings herein are especially of interest.

## Weaknesses

(1) Most minor concerns were addressed during revisions, including additional details in the methods and results sections that help interpret the data as presented.

(2) The AgRP staining is unclear. For example, in Figure 2, the figure legend says "labeled AgRP terminals (red)" (Fig 2A-D) but then concludes no difference in the number of "AgRP neurons" (Fig 2J). Is this quantification of AgRP+ neurons, terminals, or both?

(3) The PLX experiments are critical to their conclusion that during lactation, microglia in the PVN sculpt AgRP inputs; however, there is no demonstration that PLX treatment effectively eliminated microglia during this postnatal window. Microglia depletion was only assessed at P55, a month past the PLX treatment window making it unclear when and by what percentage the microglia were eliminated.

<https://doi.org/10.7554/eLife.101391.2.sa3>

**Reviewer #2 (Public review):**

Hypothalamic neural circuits that control body weight develop during the lactation period in rodents. Exposure to maternal high-fat diet during this period (MHFD-L) program has lasting effects on their neuroanatomical organization and function. Microglia sense environmental signals and can sculpt developing circuits by promoting or pruning synaptic connections. Here, the authors examine the contribution of microglia to the effects of MHFD-L to reduce projections from AgRP neurons in the ARH to the PVH, a critical node in circuits regulating energy balance. Using detailed histomorphometric analyses of Iba-1+ cells in the three brain regions (ARH, PVH, and BNST) at two time points (P16 and P30), the authors show that microglial volume and complexity increase, while cell numbers decrease across this period. Exposure to MHFD-L is associated with a transient increase in microglial complexity/volume at P16 in the PVH but not in the other brain regions or time points assessed. Depleting microglia using a pharmacological approach reversed effects of MHD-L on AgRP outgrowth and body weight.

## Strengths:

(1) The Introduction is well-written and provides a good overview of what is known about the roles of microglia in sculpting developing circuits in the hippocampus and cortex. This

provides a strong rationale for the current investigations in the hypothalamus.

(2) High-quality imaging and detailed 3-D reconstructions of Iba-1 staining in microglia are used to perform unbiased analyses of microglial complexity and to quantify the spatial relationship between microglial processes and AgRP terminals.

Weaknesses:

(1) The central claim of the manuscript is that microglia in the PVH sculpt the density of AgRP inputs to the PVH in a temporally and spatially restricted manner. While the findings of the microglial ablation experiment are consistent with this hypothesis, they do not prove causality, since their manipulations were not limited to the PVH. Further studies are needed to exclude the possibility that increased outgrowth from AgRP neurons results from direct actions in the ARH or indirect consequences of changes in growth rates.

(2) Impacts of microglial depletion were only assessed in adulthood. Given the hypothesized importance of differences in microglia at P16 and not at P30, it would be helpful to demonstrate that PLX5622 does indeed affect microglia at P16, when the circuit is most sensitive to maternal influences.

<https://doi.org/10.7554/eLife.101391.2.sa2>

### **Reviewer #3 (Public review):**

Summary:

The authors interrogated the putative role of microglia in determining AgRP fiber maturation in offspring exposed to a maternal high-fat diet. They found that changes in specific parts of the hypothalamus (but not in others) occur in microglia and that the effect of microglia on AgRP fibers appears to be beyond synaptic pruning, a classical function of these brain-resident macrophages.

Strengths:

The work is very strong in neuroanatomy. The images are clear and nicely convey the anatomical differences. The microglia depletion study adds functional relevance to the paper; however, the pitfalls of the technology regarding functional relevance should be discussed.

Weaknesses:

There was no attempt to functionally interrogate microglia in different parts of the hypothalamus. Morphology alone does not reflect a potential for significant signaling alterations that may occur within and between these and other cell types.

Comments on revised submission: My advice is to change the title by removing "required" and state what is interrogated and found in the paper. A more accurate title would be (for example): Implication of Microglia for Developmental Specification of AgRP Innervation in the Hypothalamus of Offspring Exposed to Maternal High-Fat Diet During Lactation.

I suggest that the authors discuss the limitations of their approach and findings, and propose future directions to address them

<https://doi.org/10.7554/eLife.101391.2.sa1>

## Author response:

The following is the authors' response to the original reviews

### **Public Reviews:**

#### **Reviewer #1 (Public reviews):**

*(1) A cartoon paradigm of the HFD treatment window would be a helpful addition to Figure 1. Relatedly, the authors might consider qualifying MHFD as 'lactational MHFD.' Readers might miss the fact that the exposure window starts at birth.*

This is a good suggestion. The MHFD-L model has been used previously (e.g. Vogt et al. 2014). We have included a cartoon of the MHFD-L model and the PLX treatments to Figure 4, which we feel helps the readers and thank the reviewer for the suggestion.

*(2) More details on the modeling pipeline are needed either in Figure 1 or text. Of the ~50 microglia that were counted (based on Figure 1J), were all 50 quantified for the morphological assessments? Were equal numbers used for the control and MHFD groups? Were the 3D models adjusted manually for accuracy? How much background was detected by IMARIS that was discarded? Was the user blind to the treatment group while using the pipeline? Were the microglia clustered or equally spread across the PVN?*

In response to this suggestion, we have expanded the description of the image analysis routine in the methods. The analysis focused on detailed changes in microglial morphology as opposed to overall changes in microglia throughout the PVH as a whole. Accordingly, we applied anatomically matched ROIs to the PVH for the measurements. As described in the methods, the Imaris Filaments tool was used to visualize microglia fully contained within a tissue section and a mask derived from the 3D model for these cells was used to isolate them for further analysis, thereby separating these cells from interstitial labeling corresponding to parts of cell processes or other labeling not associated with selected cells. There was no formal "background subtraction." This was an error in the previous version of the manuscript and we have revised the methods to reflect the process actually used. The images were segmented (to enhance signal to noise for 3D rendering), and then a Gaussian filter was applied to improve edge detection, which facilitates the morphological measurements.

*(3) Suggest toning back some of the language. For example: "...consistent with enhanced activity and surveillance of their immediate microenvironment" (Line 195) could be "...perhaps consistent with...". Likewise, "profound" (Lines 194, 377) might be an overstatement.*

Revisions have been made to both the Introduction and Discussion to modulate our representation of this controversial issue.

*(4) Representative images for AgRP+ cells (quantified in Figure 2J) are missing. Why not a co-label of Iba1+/AgRP+ as per Figure 1, 3? Also, what was quantified in Figure 2J - soma? Total immunoreactivity?*

Because the density of AgRP labeling does not change in the ARH we omitted the red channel image (AgRP labeling) to highlight the similarity of the microglial morphology. To address the reviewer's concerns, in the revised figure we have reconstituted the figure with both the green (microglial) and red (AgRP) channels depicted.

Figure 2J displays the numbers of AgRP neurons counted in the ARH in selected ROIs through the ARH. The Methods section has been revised to include the visualization procedure used

for the cell counts.

(5) For the PLX experiment:

a) "...we depleted microglia during the lactation period" (Line 234). This statement suggests microglia decreased from the first injection at P4 and throughout lactation, which is inaccurate. PLX5622 effects take time, upwards of a week. Thus, if PLX5622 injections started at P4, it could be P11 before the decrease in microglia numbers is stable. Moreover, by the time microglia are entirely knocked down, the pups might be supplementing some chow for milk, making it unclear how much PLX5622 they were receiving from the dam, which could also impact the rate at which microglia repopulation commences in the fetal brain. Quantifying microglia across the P4-P21 treatment window would be helpful, especially at P16, since the PVN AgRP microglia phenotypes were demonstrated and roughly when pups might start eating some chow.

b) I am surprised that ~70% of the microglia are present at P21. Does this number reflect that microglia are returning as the pups no longer receive PLX5622 from milk from the dam? Does it reflect the poor elimination of microglia in the first place?

This is an important point and have revised the first sentence in section 2.3 to clarify the PLX treatment logic and added a cartoon to Fig. 4 to show the treatment timeline. The PLX5622 was not administered to the dams but daily to the pups. We also agree with the interpretation that PLX5622 depleted numbers of microglia, as supported by the microglial cell counts, rather than effected a complete elimination and have made revisions to clarify this distinction. Although mice were weighed at weaning, cellular measurements were only made in mice perfused at P55.

(6) Was microglia morphology examined for all microglia across the PVN? It is possible that a focus on PVNmpd microglia would reveal a stronger phenotype? In Figure 4H, J, AgRP+ terminals are counted in PVN subregions - PVNmpd and PVNpml, with PVNmpd showing a decrease of ~300 AgRP+ terminals in MHFD/Veh (rescued in MHFD/PLX5622). In Figure 1K, AgRP+ terminals across what appears to be the entire PVN decrease by ~300, suggesting that PVNmpd is driving this phenotype. If true, then do microglia within the PVNmpd display this morphology phenotype?

We have revised the description of the analysis procedures to clarify these points. All measurements were made in user defined, matched regions of interest according to morphological features of the PVH. No measurements were made that included the entire PVH and we revised the Methods section to improve clarity.

(7) What chow did the pups receive as they started to consume solid food? Is this only a MHFD challenge, or could the pups be consuming HFD chow that fell into the cage?

The pups were weaned onto the same normal chow diet that the dams received prior to MHFD-L treatment. The cages were inspected daily and minimal HFD spillage was observed, although we cannot rule out with certainty any contribution of the pups directly consuming the HFD. We have edited Methods section 5.2 for clarity.

(8) Figure 5: Does internalized AgRP+ co-localize with CD68+ lysosomes? How was 'internalized' determined?

This important point has been clarified by revisions to the Methods section.

(9) Different sample sizes are used across experiments (e.g., Figure 4 NCD n=5, MHFD n=4). Does this impact statistical significance?

Sample size does impact power of ANOVA with larger samples reducing the chance of errors. ANOVA is generally robust in the face of moderate departures from the assumption of equal sample sizes and equal variance such as we experienced in the PLX5622 experiment. Here we used t-tests to detect differences in a single variable between two groups and two-way ANOVA to compare treatment by diet and treatment changes in the PLX5622 studies. Additional detail has been added to the Methods section to clarify this point.

**Reviewer #2 (Public reviews):**

*(1) Under chow-fed conditions, there is a decrease in the number of microglia in the PVH and ARH between P16 and P30, accompanied by an increase in complexity/volume. With the exception of PVH microglia at P16, this maturation process is not affected by MHFD. This "transient" increase in microglial complexity could also reflect premature maturation of the circuit.*

This is an interesting possibility that requires future investigation (see response to Recommended Suggestions, above).

*(2) The key experiment in this paper, the ablation of microglia, was presumably designed to prevent microglial expansion/activation in the PVH of MHFD pups. However, it also likely accelerates and exaggerates the decrease in cell number during normal development regardless of maternal diet. Efforts to interpret these findings are further complicated because microglial and AgRP neuronal phenotypes were not assessed at earlier time points when the circuit is most sensitive to maternal influences.*

We agree that evaluations of microglia and hypothalamic circuits at many more time points would indeed be informative (see comments above).

*(3) Microglial loss was induced broadly in the forebrain. Enhanced AgRP outgrowth to the PVH could be caused by actions elsewhere, such as direct effects on AgRP neurons in the ARH or secondary effects of changes in growth rates.*

A local effect of microglia in the ARH that affects growth of AgRP axons remains a distinct possibility that deserves a targeted examination (see response to Recommended Suggestions, above).

*(4) Prior publications from the authors and other groups support the idea that the density of AgRP projections to the PVH is primarily driven by factors regulating outgrowth and not pruning. The failure to observe increased engulfment of AgRP fibers by PVH microglia is therefore not surprising. The possibility that synaptic connectivity is modulated by microglia was not explored.*

Synaptic pruning and regulation of axon targeting are not mutually exclusive processes and microglia may participate in both. Here we evaluated innervation of the PVH, which is sensitive to MHFD-L exposure, and engulfment of AgRP terminals by microglia, which does appear to be altered by MHFD-L. Given previous observations of terminal engulfment by microglia in other brain regions in response to environmental changes (e.g. prolonged stress) it is not unreasonable to expect this outcome in the offspring of MHFD-L dams. In future work it will be important to profile multiple cell types in the PVH for microglial dependent and MHFDL-sensitive changes in targeting of AgRP axons. Equally important is a full characterization of postsynaptic changes in PVH neurons.



**Reviewer #3 (Public reviews):**

*There was no attempt to interrogate microglia in different parts of the hypothalamus functionally. Morphology alone does not reflect a potential for significant signaling alterations that may occur within and between these and other cell types.*

*The authors should discuss the limitations of their approach and findings and propose future directions to address them.*

We agree that evaluations of microglia and hypothalamic circuits at many more time points that include analyses of multiple regions would indeed be informative. We have added statements to the manuscript that address the limitations of our experimental approach and suggest future studies that will extend understanding of underlying mechanisms beyond those investigated here.

**Recommendations for the authors:****Reviewing Editors Comments:**

*(1) The Abstract is 405 words and should be shortened to less than 200 words.*

The abstract has been edited to 200 words.

*(2) The authors might consider raising the question in the Introduction of whether reduced AgRP innervation of the PVN in MHFD-treated mice is due to decreased axonal growth, enhanced microglial-mediated pruning, or a combination of both. The potential effects on axonal growth should be given more consideration.*

This is an important point that we agree deserves additional consideration in the manuscript. Our past work has focused on leptin's ability to influence axonal targeting of PVH neurons by AgRP and PPG neurons through a cell-autonomous mechanism and our conclusion is that leptin primarily induces axon growth. Because in this study our design did not focus on changes in axon growth over time but on regional changes in microglia and their interactions with AgRP terminals we did not want to divert attention from our logic in the introduction by highlighting multiple mechanisms. However, we have added a brief mention in the Introduction and have expanded consideration of axonal growth effects to the Discussion. Distinguishing between microglia's role in synaptic density or axon targeting in this pathway is an important goal of future work.

*(3) Line 37, a high-fat diet should be defined here as HFD and used consistently thereafter. Note that "high-fat-diet exposure" requires two hyphens.*

The suggested revisions have been made throughout the manuscript.

*(4) Line 38 and elsewhere, MHFD does not adequately describe the treatment being limited to the lactation period, perhaps MLHFD would be better or just LHFD (because the pups can't lactate).*

The suggested revisions have been made throughout the manuscript, and we have used MHFD-L to describe maternal consumption of a high-fat diet that is restricted to the lactation period.

*(5) Line 110, leptin-deficient mice (add hyphen).*

*(6) Line 183, NCD should be defined.*

The suggested revisions have been made throughout the manuscript.

(7) Lines 237- 238, it is not clear what is widespread in the rostral forebrain. Is it the loss of microglia? What is the dividing point between the rostral and caudal forebrain? Were microglia depleted in the caudal forebrain too?

We have revised this section of the manuscript to focus the description on the hypothalamus alone and specify that the reduction in microglial density is not restricted to the PVH.

(8) Line 245, *microglial-mediated effects* (add hyphen).

(9) Line 247, *vehicle-treated mice* (add hyphen).

The suggested revisions have been made throughout the manuscript.

(10) Line 457, when referring to genes, the approved gene name should be used in italics, *AgRP* should be *Agrp* (italics).

The suggested revision has been made throughout the manuscript.

(11) Line 459, the name of the Syn-Tom mice in the Key Resource table, Methods, and Text should be consistent. It would be best to use the formal name of the Ai34 line of mice on the JAX website.

The suggested revisions have been made throughout the manuscript.

(12) Figure 1G H, and I um should have Greek micro; Fig. 1J and K, Replace # with Number. The same suggestions apply to all the other figures.

Both the manuscript and figures have been revised in accordance with this recommendation.

(13) Figures 4 G, H, I and J. and Figures 5 M and O. The font size is too small to see well.

Fonts have been changed in the figures to improve visibility.

#### **Reviewer #1 (Recommendations for the authors):**

(1) Figures are out of order in the text. For example, Figure 1A is followed next by the results for Figure 1J instead of Figure 1B.

We regret that the organization of figure panels makes for awkward matching for the reader as they proceed through the text. We designed the figures to facilitate comparisons between cellular responses and differences in labeling. After evaluating a reorganization, we decided to maintain the original panel configurations, but have revised the text to more closely follow the presentation of cellular features in the figures.

(2) Figure 1B.: All images lack scale bars.

(3) Line 433 - 'underlie' is spelled wrong.

(4) Rosin et al should be 2019 and not 2018.

These corrections have been implemented in the revised text and figures.

(5) The statement that "the effects of MHFD on microglial morphology in the PVH of offspring display both temporal and regional specificity, which correspond to a decrease in the density of AgRP inputs to the PVH" (Line 196) needs clarification, as the phrase "regional specificity" has not been substantiated in this section even though it is discussed later.

We agree with this comment and have revised section 2.1 to more closely match the data presented to this point in the manuscript.

**Reviewer #2 (Recommendations for the authors):**

(1) The claim of "spatial specificity" in the effects of MHFD on microglia is based on an increase in the complexity/volume of microglia at P16 in the PVH that was not seen in the ARH or BNST. The transient nature of the effect raises several questions: Does the effect on the PVH represent premature maturation?

This is an interesting suggestion. However, given how little is known about microglial maturation in the hypothalamus it is difficult to address. It is indeed possible that microglia mature at different rates in each AgRP target, and that MHFD-L exposure alters the rate of maturation in some regions but not others. This will require a great deal more analysis of both microglia and ARH projections to understand fully (see below).

(2) To support their central claim that microglia in the PVH "sculpt the density of AgRP inputs to the PVH" the authors report effects on Iba1+ cells in the PVH of chow-fed dams at P55, body weight at P21, and AgRP projections in the PVH at an unspecified age. It is hard to understand what is happening across "normal" development in chow-fed dams since the number of Iba1+ cells decreases from ~50 to ~25 between P16 and P30 (Figure 1), but then increases to >60 cells at P55 (Figure 4). Given the large fluctuations in microglial population across time, analyzing the same parameters (i.e. microglial number/morphology in the ARH and PVH, AgRP neuronal number in the ARH, and fiber density in the PVH, and body weight) across time points before, during and after the critical period in chow and MHFD conditions would be very helpful.

The time points we evaluated were chosen to be during and after the previously determined critical period for development of AgRP projections to the PVH, which were then compared with adults (which were all P55) to assess longevity of the effects. We have incorporated revisions to improve the clarity of when measurements were assessed, and treatments implemented. Defining the cellular dynamics of microglia across time remains a major challenge for the field and will certainly be informed by future studies with additional time points, as well as by in vivo imaging studies focused on regions identified here. Although such studies are beyond the scope of the present work, their completion would advance our current understanding of how microglia respond to nutritional changes during development of feeding circuits.

(3) As microglia are also ablated in the ARH, direct effects on AgRP neurons or indirect effects via changes in growth rates could also contribute to increased AgRP fiber density in the PVH. In support of the first possibility, postnatal microglial depletion increases the number of AgRP neurons (Sun, et al. 2023).

We agree with the suggestion, also raised by the Reviewing Editor, which has been addressed briefly in the Introduction, and in more detail by revisions to the Discussion section.

*(4) The failure to assess alpha-MSH fibers in the same animals was a missed opportunity. They are also affected by MHFD but likely involve a distinct mechanism (Vogt, et al 2014).*

Given the paired interest in POMC neurons and AgRP neurons I understand the reviewer's comment. We chose to focus solely on AgRP neurons because we do not currently have a way to genetically target axonal labeling exclusively to POMC neurons due to the shared precursor origin of POMC neurons and a percentage of NPY neurons in the ARH, as shown by Lori Zeltser's laboratory. Moreover, the elegant work by Vogt et al. focused on responses of POMC neurons in the MHFD-L model. However, it certainly remains possible that microglia in the PVH interact with terminals derived from POMC neurons, as well as with terminals derived from other afferent populations of neurons.

*(5) All statistical analyses involved unpaired t-tests. Two-way ANOVAs should be used to assess the effects of age and HFD and interactions between these factors.*

We used t-tests to detect differences in a single variable between two groups and two-way ANOVA to compare treatment by diet and treatment changes in the PLX5622 studies. Additional detail has been added to the Methods section and information added to the figure legend for Fig. 4 to clarify this point.

**Reviewer #3 (Recommendations for the authors):**

*I suggest exploring the deeper characterization of the microglia in various parts of the hypothalamus in different conditions. This could include cytokine assessment or spatial transcriptomic.*

We agree that a great deal more work is needed to improve our understanding of how microglia impact hypothalamic development more broadly and to identify underlying molecular mechanisms. We are hopeful that the data presented here will motivate additional study of microglial dynamics in multiple hypothalamic regions, as well as detailed studies of cellular signaling events for factors derived from MHFD-L dams that impact neural development in the hypothalamus.

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